



Adverse Clinical Signs in Neuromedical and Neurosurgical Patients

In addition to NEWS triggers

ADVERSE SIGN

A & B

- Urgent airway management required if:
 - GCS drops to **8 or less**
 - New or **prolonged** (>5 minutes) seizure
- Oxygen saturation $\leq 91\%$ on room air
- Respiratory rate ≤ 8 or $\geq 25/\text{min}$ (1)
- Vital capacity $< 15\text{ml/kg}$

C

- Heart rate ≤ 40 or $\geq 130\text{bpm}$ (1)
- Systolic BP ≤ 100 or $\geq 180\text{mmHg}$ (2)
- In **SAH** (unprotected aneurysm) SBP trigger $> 160\text{mmHg}$
- In **acute ICH** SBP trigger $> 140\text{mmHg}$

- (1) Patients with brain injury may have individual HR & RR limits
(2) Patients with autonomic nervous system dysfunction may have individual HR & SBP limits

D

Glasgow Coma Scale

- Drop of ≥ 2 points **immediate** escalation / review
- Drop of **1** point for $> 30\text{min}$ escalation / review

Limb Assessments in Spinal Patients

- Decrease in power or sensation (Spinal Obs Chart)

Seizure activity

Non-epileptic & all neurosurgical patients

- New seizure of any form

Patients with known epilepsy

- Seizure lasting > 5 minutes
- Repeated seizure within 1 hour of 1st seizure
- Failure to be orientated in time / place (or back to baseline) within 15 minutes of seizure

A drop in GCS to 8 or less or prolonged seizure (>5 minutes) also requires Anaesthetic SpR and Outreach Team to attend **immediately**

E

- Temperature $\leq 35^\circ\text{C}$
- Temperature $\geq 39^\circ\text{C}$
- $\text{Na}^+ < 125$ or $> 150\text{mmol/l}$
- $\text{K}^+ > 6\text{mmol/l}$
- Glucose ≤ 3 or $\geq 20\text{mmol/l}$
- Poorly controlled pain
- Urine output $< 100\text{ml}$ or $> 1000\text{ml}$ in 4 hours
- 'Worried about patient'

Autonomic Dysreflexia

Sudden & potentially lethal surge in BP with spinal cord lesion at or above T6 - Often triggered by acute pain or noxious stimulus - patient may normally have very low BP $\therefore$ 'normal' BP may represent significant rise

Acute kidney injury

Increased risk with certain drugs / CT contrast - $\uparrow$ serum creatinine $\geq 26\text{mmol/l}$ in 48h - $\downarrow$ urine output

Acute Abdomen

Increased risk in spinal patient

Misplaced Nasogastric Tube

Observe for **CEASE** signs

Complications of Anterior Neck Surgery

Observe for **SHOUT** signs

Cerebral Vasospasm after SAH

Observe for **SAVE** signs

Acute Vascular Bleeding

Post-Endovascular Intervention / Angiography via groin puncture / Removal of Vascath
Observe for **ESCALATE** signs

Post carotid endarterectomy

WARNING

HIGH RISK

MEDIUM RISK

ASSESSMENT

- Airway
- Breathing
- Circulation
- Disability
- Exposure
- Blood Glucose

- Patient assessed immediately by nurse-in-charge who must instigate Escalation Process
- Increase frequency of observations
- Facilitate assessment with monitoring (ECG, NIBP, SpO₂) - Monitoring mandatory for all patient transfers (to CT scan, Xray, ITU, etc.)

Troubleshooting tips while awaiting review:

If SpO₂ ↓

- Reposition patient and sit up if allowed
- Give oxygen therapy
- If tracheostomy patient, follow emergency procedure

If BP / urine output ↓

- Consider fluid bolus 100ml IV +/- repeat
- Consider bladder washout if urine output ↓

If urine output ↑ check specific gravity

If ↑ temperature consider sepsis

- Send cultures according to policy
- Consider IV fluids

Seizure management

- Seizure may self-terminate & require only supportive treatment

Supportive treatment

- Turn patient on side to protect airway
- O₂ via non-rebreather mask 12-15L/min
- Make environment safe

Drugs

- Lorazepam 0.07mg/kg at $< 2\text{mg/min}$
- Slow IV bolus 4mg into large vein, repeated if required after 10min
- Consider lower dose if small, frail or elderly
- or Diazepam IV (maximum rate of 5mg/min)
- or Midazolam IM 7.5mg or oromucosal solution either 5mg/1ml or 10mg/2ml

Autonomic Dysreflexia Signs / Symptoms

- Pounding headache
- Flushed skin above level of lesion
- Nasal congestion

Treatment – must be initiated quickly

- Sit patient upright
- Identify & correct noxious stimulus
- May require urgent medication to reduce BP
- Record BP every 5min until episode resolves

Acute kidney injury

- ABCDE assessment
- If NEWS triggering give oxygen
- Maintenance fluids = output + 500ml
- Fluid chart & daily weight
- Review drug dosages
- Daily bloods – urea, electrolytes, liver, bone

Acute Abdomen

- ABCDE assessment
- Assess for abdominal distension / pain
- Date bowels last opened

ESCALATION PROCESS

If patient triggers **one or more** Adverse Clinical Sign
Nurse-in-charge to assess patient and follow Escalation Process below

MEDIUM RISK

Nurse-in-charge to assess patient and inform Neurology or Neurosurgery SHO

SHO must attend within **15 minutes**, document findings and plan of action in medical notes and discuss patient with SpR

If SHO fails to respond / attend within **15 minutes** contact SHO again

If no response within **5 minutes** contact SpR

If SpR does not respond within **5 minutes** contact Consultant on call

HIGH RISK

Nurse-in-charge to assess patient and request Neurology or Neurosurgery SHO to attend **immediately**

SHO must attend **immediately**, document findings and plan of action in medical notes and discuss patient with SpR

If SHO fails to respond / attend **immediately** contact SHO again

If no **immediate** response contact SpR

If SpR does not respond **immediately** contact Consultant on call



The following people **must** be contacted (Refer patient using SBAR)

Situation
Background
Assessment
Recommendation



Neurosurgery / Neurology Team
Outreach Team: Bleep 8277

+/- Anaesthetic SpR

1st on call Bleep 8131

2nd on call Bleep 8011

In emergency call 2222

+ If required call Clinical Site Practitioner: Bleep 8240

ALERTS

! This patient has had anterior neck surgery!

- Swelling of neck
 - Hoarseness / voice changes
 - Oesophageal dysphagia / droo
 - Unusual behaviour / agitation
 - Tachypnoea / difficulty breathing
- If any of these symptoms are present Neurosurgical SpR immediate

CAROTID ENDARTERECTOMY

- Wound / neck swelling / bleeding
- Airway compromise / breathing difficulties / tachypnoea / hoarseness / voice changes
- Rhythm changes / ECG changes
- Neurological deficit
- Increasing / decreasing parameters
- New or increasing tachypnoea
- Get immediate SpR signs or symptoms

MONITORING FOR CEREBRAL VASOSPASM AFTER ANEURYSMAL SUBARACHNOID HAEMORRHAGE (SAH)

- 'SAVE' signs or symptoms may be present without a change in GCS
- Severe persisting headache resistant to analgesia
- Subtle changes in speech - Is patient able to name objects?
- Arm drift - Is patient able to equally maintain both arms outstretched, palms upwards with eyes closed?
- Visual acuity - Is patient able to read as well as prior to SAH?
- Elevated BP - Is BP elevated?
- Neurological signs or symptoms
- If any 'SAVE' sign or symptom present inform nurse in charge / medical team immediately & increase frequency of observations
- Also escalate if patient has a change in behaviour, such as **AGITATION, AGITATION or CONFUSION**
- Monitor more closely if patient's relatives report any change

ESCALATION PROTOCOL	
Post-Endovascular Intervention / Angiogram via GROIN Puncture or Post Insertion / Removal of a Vascular Access Device (e.g. Vascath)	
ESCALATE	IF CT SHOWS ABNORMAL BLEEDING ACTIVATE 'PATHWAY FOR ACUTE VASCULAR BLEEDING'
Escalate alert activated if routine post-procedure observations identify possible complications	Neurosurgery SpR to make immediate contact with those responsible for patient care
Endovascular treatment / angiogram with groin puncture or insertion / removal of a vascular access device	Neurosurgery SpR to make immediate contact with those responsible for patient care
Any of the following signs / symptoms present?	Neurosurgery SpR to make immediate contact with those responsible for patient care
Cardiovascular changes - hypotension, tachycardia, cool extremities	Neurosurgery SpR to make immediate contact with those responsible for patient care
Abdominal pain / groin pain	Neurosurgery SpR to make immediate contact with those responsible for patient care
Agitation / Drop in GCS	Neurosurgery SpR to make immediate contact with those responsible for patient care
Saturation drop / Increasing O ₂ requirement	Neurosurgery SpR to make immediate contact with those responsible for patient care
Elevated heart rate / Chest pain	Neurosurgery SpR to make immediate contact with those responsible for patient care
If any CEASE symptoms are present (especially following recent insertion of NG tube) inform nurse in charge immediately. Refer to UCLH Policy Nasogastric tube insertion and management (ratified)	
ON GOING BLEEDING REQUIRING VASCULAR SURGICAL INPUT	
1. Contact on call Vascular Surgeon	
2. Call 2 nd on call Vascular Surgeon	
3. Call Clinical Director or Deputy Ext 32405	
IF PATIENT ACCEPTED FOR TRANSFER: Neurosurgery SpR to make immediate contact with those responsible for patient care	
IF PATIENT TOO UNSTABLE FOR TRANSFER: Neurosurgery SpR to make immediate contact with those responsible for patient care	